

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **alpha-Methylstyrene, stabilized**  
 Product Description: **alpha-Methylstyrene, stabilized**  
 Cat No. : 127710000; 127710010; 127710100; 127711000; 127710025  
 Synonyms Isopropenylbenzene; 2-Phenyl-1-propene  
 CAS No 98-83-9  
 Molecular Formula C9 H10

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Flammable liquids                                  | Category 3 (H226) |
| Aspiration Toxicity                                | Category 1 (H304) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Skin Sensitization                                 | Category 1 (H317) |
| Reproductive Toxicity                              | Category 2 (H361) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |
| Chronic aquatic toxicity                           | Category 2 (H411) |

**Label Elements**


# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

## Signal Word

## Danger

### Hazard Statements

H226 - Flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H361 - Suspected of damaging fertility or the unborn child  
H411 - Toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Lachrymator (substance which increases the flow of tears)  
Toxicity to Soil Dwelling Organisms  
Contains a known or suspected endocrine disruptor  
Contains a substance on the National Authorities Endocrine Disruptor Lists

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component             | CAS No  | Weight % |
|-----------------------|---------|----------|
| 4-tert-Butyl catechol | 98-29-3 | 0.0015   |
| alfa-Methylstyrene    | 98-83-9 | >95      |

## SECTION 4: FIRST AID MEASURES

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

## Description of first aid measures

### **General Advice**

If symptoms persist, call a physician.

### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

### **Ingestion**

Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

### **Self-Protection of the First Aider**

Use personal protective equipment as required.

## Most important symptoms and effects, both acute and delayed

May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## Indication of any immediate medical attention and special treatment needed

### **Notes to Physician**

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Flammable. Vapors may travel to source of ignition and flash back. Vapors may form explosive mixture with air. Containers may explode when heated. Vapors may form explosive mixtures with air.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

## Environmental precautions

Do not flush into surface water or sanitary sewer system.

## Methods and Material for Containment and Cleaning Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Keep container tightly closed in a dry and well-ventilated place. Keep refrigerated.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component          | Malaysia | ACGIH TLV   | OSHA PEL                                                                                                                                                                                |
|--------------------|----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alfa-Methylstyrene |          | TWA: 10 ppm | (Vacated) TWA: 50 ppm<br>(Vacated) TWA: 240 mg/m <sup>3</sup><br>Ceiling: 100 ppm<br>Ceiling: 480 mg/m <sup>3</sup><br>(Vacated) STEL: 100 ppm<br>(Vacated) STEL: 485 mg/m <sup>3</sup> |

| Component          | European Union                                                                                                      | The United Kingdom                                                                                                | Germany                                                                                                                                                                                                                                                            |
|--------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| alfa-Methylstyrene | TWA: 50 ppm (8h)<br>TWA: 246 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 492 mg/m <sup>3</sup> (15min) | STEL: 100 ppm 15 min<br>STEL: 491 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 246 mg/m <sup>3</sup> 8 hr | TWA: 50 ppm (8 Stunden). AGW -<br>exposure factor 2<br>TWA: 250 mg/m <sup>3</sup> (8 Stunden). AGW<br>- exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 250 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 500 mg/m <sup>3</sup> |

### Exposure Controls

#### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

## Personal protective equipment

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                          |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

|                                |                                                                       |
|--------------------------------|-----------------------------------------------------------------------|
| <b><u>Hygiene Measures</u></b> | Handle in accordance with good industrial hygiene and safety practice |
|--------------------------------|-----------------------------------------------------------------------|

|                                               |                                                                                               |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b><u>Environmental exposure controls</u></b> | Prevent product from entering drains Do not allow material to contaminate ground water system |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------|

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                                                |                                          |
|-------------------------------------|------------------------------------------------|------------------------------------------|
| <b>Appearance</b>                   | Colorless                                      |                                          |
| <b>Physical State</b>               | Liquid                                         |                                          |
| <b>Odor</b>                         | aromatic                                       |                                          |
| <b>Odor Threshold</b>               | No data available                              |                                          |
| <b>pH</b>                           | 5-6                                            | 500 g/l aq.sol                           |
| <b>Melting Point/Range</b>          | -23 °C / -9.4 °F                               |                                          |
| <b>Softening Point</b>              | No data available                              |                                          |
| <b>Boiling Point/Range</b>          | 165 - 169 °C / 329 - 336.2 °F                  | @ 760 mmHg                               |
| <b>Flash Point</b>                  | 45 °C / 113 °F                                 | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | No data available                              |                                          |
| <b>Flammability (solid,gas)</b>     | Not applicable                                 | Liquid                                   |
| <b>Explosion Limits</b>             | <b>Lower</b> 0.9 Vol%<br><b>Upper</b> 6.6 Vol% |                                          |
| <b>Vapor Pressure</b>               | 2.9 mbar @ 20 °C                               |                                          |
| <b>Vapor Density</b>                | 4.1 (Air = 1.0)                                | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>   | 0.909                                          |                                          |
| <b>Bulk Density</b>                 | Not applicable                                 | Liquid                                   |
| <b>Water Solubility</b>             | Insoluble                                      |                                          |
| <b>Solubility in other solvents</b> | No information available                       |                                          |

**Partition Coefficient (n-octanol/water)**

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

|                                  |                          |                                        |
|----------------------------------|--------------------------|----------------------------------------|
| <b>Component</b>                 | <b>log Pow</b>           |                                        |
| 4-tert-Butyl catechol            | 1.98                     |                                        |
| alfa-Methylstyrene               | 3.48                     |                                        |
| <b>Autoignition Temperature</b>  | 445 °C / 833 °F          |                                        |
| <b>Decomposition Temperature</b> | No data available        |                                        |
| <b>Viscosity</b>                 | 0.94 cP at 20 °C         |                                        |
| <b>Explosive Properties</b>      |                          | explosive air/vapour mixtures possible |
| <b>Oxidizing Properties</b>      | No information available |                                        |
| <b>Molecular Formula</b>         | C9 H10                   |                                        |
| <b>Molecular Weight</b>          | 118.18                   |                                        |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under recommended storage conditions.

### Possibility of Hazardous Reactions

|                                 |                                     |
|---------------------------------|-------------------------------------|
| <b>Hazardous Polymerization</b> | Hazardous polymerization may occur. |
| <b>Hazardous Reactions</b>      | None under normal processing.       |

### Conditions to Avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

### Incompatible Materials

Acids. Strong oxidizing agents. Finely powdered metals. Peroxides.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Based on available data, the classification criteria are not met |
| Dermal              | Based on available data, the classification criteria are not met |
| Inhalation          | Based on available data, the classification criteria are not met |

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

| Component             | LD50 Oral                 | LD50 Dermal          | LC50 Inhalation     |
|-----------------------|---------------------------|----------------------|---------------------|
| 4-tert-Butyl catechol | 815 mg/kg ( Rat )         | 1331 mg/kg ( Rat )   | -                   |
| alfa-Methylstyrene    | LD50 = 4900 mg/kg ( Rat ) | 14560 mg/kg (Rabbit) | 22.85 mg/L/6h (Rat) |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

Category 1

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component          | EU | UK | Germany | IARC     |
|--------------------|----|----|---------|----------|
| alfa-Methylstyrene |    |    |         | Group 2B |

(g) reproductive toxicity;  
Reproductive Effects

Category 2

Contains ingredients that are suspected reproductive hazards.

(h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

Category 1

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

Endocrine Disrupting Properties  
Assess endocrine disrupting  
properties for human health

Contains a substance on the National Authorities Endocrine Disruptor Lists

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component             | Freshwater Fish      | Water Flea         | Freshwater Algae | Microtox |
|-----------------------|----------------------|--------------------|------------------|----------|
| 4-tert-Butyl catechol | LC50 = 0.12 mg/L 96h | EC50=0.48 mg/L 48h |                  |          |

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

|                    |                                                                                     |                      |  |  |
|--------------------|-------------------------------------------------------------------------------------|----------------------|--|--|
| alfa-Methylstyrene | LC50: 28 mg/L/48h<br>(Leuciscus idus)<br>LC50: 2,97 mg/L/96h<br>(Brachydanio rerio) | EC50: 1,645 mg/L/48h |  |  |
|--------------------|-------------------------------------------------------------------------------------|----------------------|--|--|

## Persistence and degradability

**Persistence**  
**Degradation in sewage treatment plant**

Persistence is unlikely.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component             | log Pow | Bioconcentration factor (BCF) |
|-----------------------|---------|-------------------------------|
| 4-tert-Butyl catechol | 1.98    | No data available             |
| alfa-Methylstyrene    | 3.48    | 15 - 140 dimensionless        |

## Mobility in soil

Spillage unlikely to penetrate soil. The product is insoluble and floats on water. The product evaporates slowly. . Is not likely mobile in the environment due its low water solubility.  
Spillage unlikely to penetrate soil.

## Endocrine Disruptor Information **Assess endocrine disrupting properties for the environment**

This product does not contain any known or suspected endocrine disruptors  
Contains a substance on the National Authorities Endocrine Disruptor Lists.

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not let this chemical enter the environment Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN2303  
Hazard Class 3  
Packing Group III  
Proper Shipping Name ISOPROPENYLBENZENE

### Road and Rail Transport

UN-No UN2303  
Hazard Class 3  
Packing Group III

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

Proper Shipping Name ISOPROPENYLBENZENE

## IATA

UN-No UN2303  
Hazard Class 3  
Packing Group III  
Proper Shipping Name ISOPROPENYLBENZENE

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component             | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| 4-tert-Butyl catechol | 202-653-9 | X    | X   | X     | X    | X    | X     | X    | KE-11368 |
| alfa-Methylstyrene    | 202-705-0 | X    | X   | X     | X    | X    | X     | X    | KE-23939 |

### National Regulations

Persistent Organic Pollutant This product does not contain any known or suspected substance  
Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

# SAFETY DATA SHEET

alpha-Methylstyrene, stabilized

Revision Date 21-Mar-2025

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date 21-Mar-2025  
Revision Summary Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**